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CENTRE
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CHEMISTRY

Paper 2

Grade 12

May 2023

2 hours

Candidates answer on the Question Paper.

Additional Materials: Calculator
 Ruler
 Data Booklet

12CHEM/02

READ THESE INSTRUCTIONS FIRST

Write your centre number and candidate number in the spaces at the top of the page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO NOT WRITE IN ANY BARCODES.

Answer **all** questions.

You may receive fewer marks if you do not show your working or do not use appropriate units where required.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 100.

Answer all questions in English.

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8	
9	
10	
Total	

This document consists of **16** printed pages.

1 Sulfur hexafluoride, SF_6 , is used in lung radiography.

- (a) (i) Draw the dot-and-cross diagram of the sulfur hexafluoride, SF_6 molecule (show only the outer shell electrons).

[2]

- (ii) Predict shape and bond angles of sulfur hexafluoride.

shape [1]

bond angle [1]

- (b) (i) The bonding in a sulfur hexafluoride, SF_6 molecule is covalent.

Explain the meaning of a covalent bond by using the SF_6 molecule.

.....

..... [1]

- (ii) The covalent bonds between atoms in the SF_6 molecule are covalent *polar* bonds.

Explain why the molecule is non-polar.

.....

..... [1]

- (c) Many metal ions form complex ions, which have the same shape as a SF_6 molecule. The complex ion, $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ is formed when water ligands surround Co^{2+} ion.

- (i) Explain the meaning of the term *ligand*.

.....

..... [1]

(ii) Name the type of bond formed between the water ligands and Co^{2+} .

..... [1]

(iii) Cobalt chloride paper is used in the laboratory to test the water. Sky blue cobalt chloride paper contains the complex ion $[\text{Co}(\text{Cl}_4)]^{2-}$, when water is added the pink coloured $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ complex is formed.

Explain why the colour changes.

.....

..... [1]

(iv) Write the equation for the formation of $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ from the cobalt chloride paper.

Name this type of reaction.

equation [1]

type of reaction [1]

[Total: 11]

2 The trends in the Periodic Table help explain the properties of a particular chemical element.

(a) (i) Define the term for the first ionisation energy.

.....

 [2]

(ii) Explain why the first ionisation energy of potassium is smaller than the first ionisation energy of bromine.

.....

 [2]

(b) (i) A potassium ion, K^+ , and a calcium ion, Ca^{2+} , both have the same number of electrons.

Explain why the ionic radius of potassium ion is larger.

.....
 [1]

(ii) Describe the change in *atomic radii* across Period 3.

Explain your answer.

.....

 [3]

[Total: 8]

- 3 Nowadays, fossil fuels are used to obtain heat. One of these fossil fuels is coal. Coal contains a large percentage by mass of carbon.

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- (a) (i) The complete combustion of coal is a redox reaction.

Write a balanced equation and show the oxidation number change for each element in the equation.

[3]

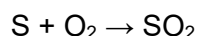
- (ii) Identify the oxidising agent and the reducing agent.

oxidising agent

reducing agent

[1]

- (b) A coal-fired power station uses a fuel that contains 0.5 % sulfur by mass.



Calculate the volume of sulfur dioxide, in dm^3 and measured at RTP, released by burning 20 tonnes of the fuel.

volume = dm^3 [4]

- (c) Sulfur dioxide is toxic and is oxidised in the air to form sulfur trioxide.

Give **two** possible environmental problems caused by sulfur trioxide.

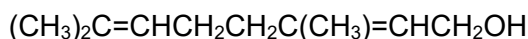
1 [1]

2 [1]

[Total 10]

- 4 Geraniol and acidified potassium dichromate (IV) completely changes the odour of the compound.

The structural formula for geraniol is shown.



- (a) (i) Geraniol and acidified potassium dichromate are heated together and the product is distilled from the reaction mixture as it is formed.

Write a balanced equation for this oxidation of geraniol.

All organic compounds should be shown as their structural formulae.

[1]

- (ii) Describe the colour change observed during the reaction.

[1]

- (b) It is possible to produce a carboxylic acid from geraniol.

State the condition needed for this reaction.

Write a balanced equation for this reaction.

All organic compounds should be shown as their structural formulae.

conditions

[1]

equation

[1]

- (c) Geraniol reacts with a small amount of cold acidified dilute potassium manganate (VII).

- (i) State which functional group will react in this reaction.

[1]

- (ii) State how many hydroxyl groups are present in the organic product of this reaction.

[1]

(d) State what type of stereoisomerism is in the product

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.....

Explain your answer.

.....

.....

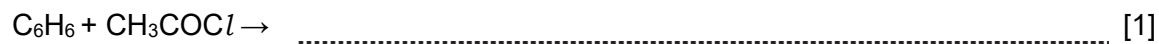
.....

[2]

[Total 8]

5 Benzene reacts with ethanoyl chloride in a Friedel-Crafts acylation.

(a) Complete the equation for the reaction between C_6H_6 and CH_3COCl .



Write the names of the products of the reaction.

..... [1]

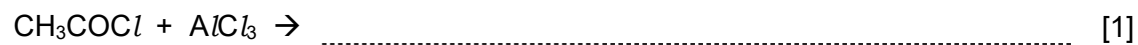
(b) The reaction between benzene and ethanoyl chloride uses an aluminium chloride catalyst.

Name the type of reaction and state one **other** reaction condition.

reaction type [1]

condition [1]

(c) Complete the equation for the reaction between ethanoyl chloride and aluminium chloride:



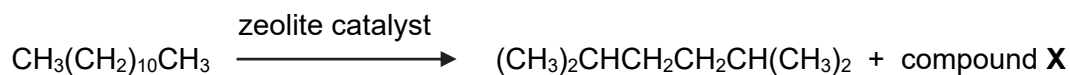
(d) Draw the mechanism for the reaction in 5(a).

[4]

[Total 9]

- 6 Dodecane, C₁₂H₂₆, is found in the kerosene fraction of crude oil.

The catalytic cracking of a dodecane molecule gives one branched alkane and one straight chain alkene molecule.



- (a) Catalytic cracking involves bond breaking known as heterolytic fission. Heterolytic fission produces positive and negative ions.

- (i) Define the term catalytic cracking.

.....
 [1]

- (ii) Write the structural formula of the compound X.

[1]

- (iii) Write an equation to show an example of heterolytic fission of a dodecane molecule.

Include electrons in the structures of the products of the fission.

[2]

- (b) Explain the industrial importance of cracking.

.....

 [2]

- (c) The main difference between *thermal* cracking and *catalytic* cracking is the way in which covalent bonds in the hydrocarbon chains are broken.

Explain the meaning of *homolytic fission*.

.....
 [1]

[Total: 7]

- 7 A range of different analytical techniques such as NMR, IR spectroscopy and mass spectrometry are used to detect and identify substances.

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(a) Look at the infrared spectrum of ethanol in Fig. 7.1.

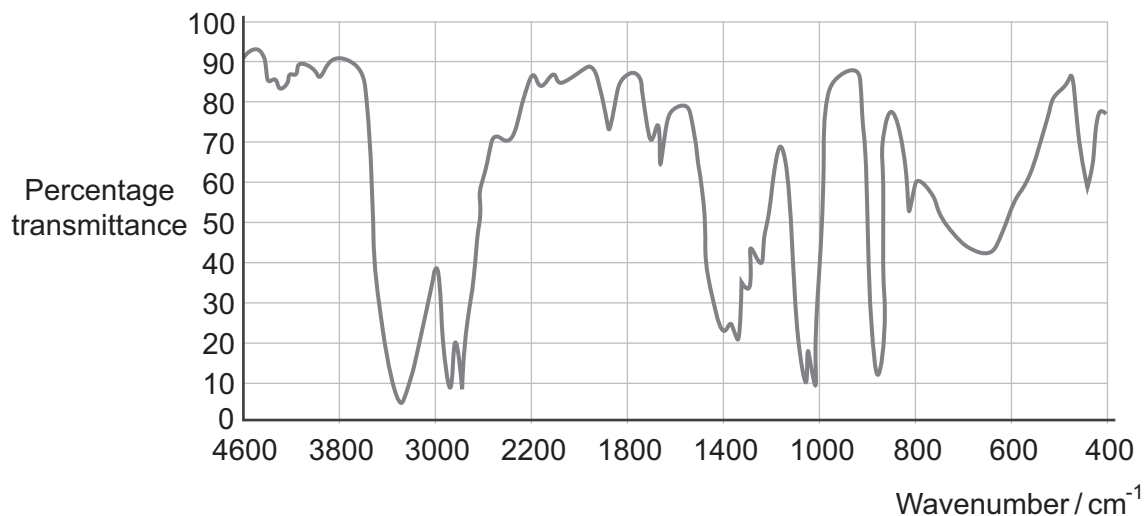


Fig. 7.1.

- (i) Identify the functional group responsible for the broad absorption trough at about 3400 cm^{-1} .

..... [1]

- (ii) Human breath contains water vapor.

Explain why the absorption range at 3400 cm^{-1} cannot be used to measure the concentration of ethanol in human breath?

.....
..... [1]

(b) Look at the high-resolution ^1H NMR spectrum of ethanol.

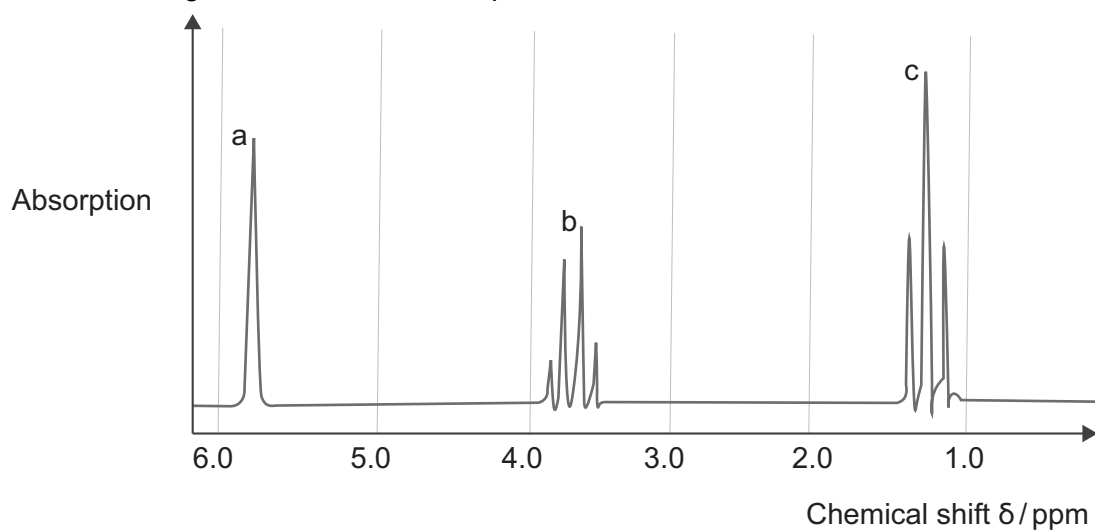


Fig. 7.2.

- (i) Define the parts of the ethanol molecule with hydrogen atoms corresponding to the peaks:

Peak a

Peak b

Peak c

[2]

- (ii) Explain the meaning of splitting in ^1H NMR spectroscopy using the $n+1$ rule.

.....

..... [1]

- (c) Heavy water, D_2O , is used to identify functional groups with labile protons.

Describe the change in the high resolution ^1H NMR spectrum of ethanol when it has D_2O added.

..... [1]

- (d) Look at the mass spectrum of ethanol.

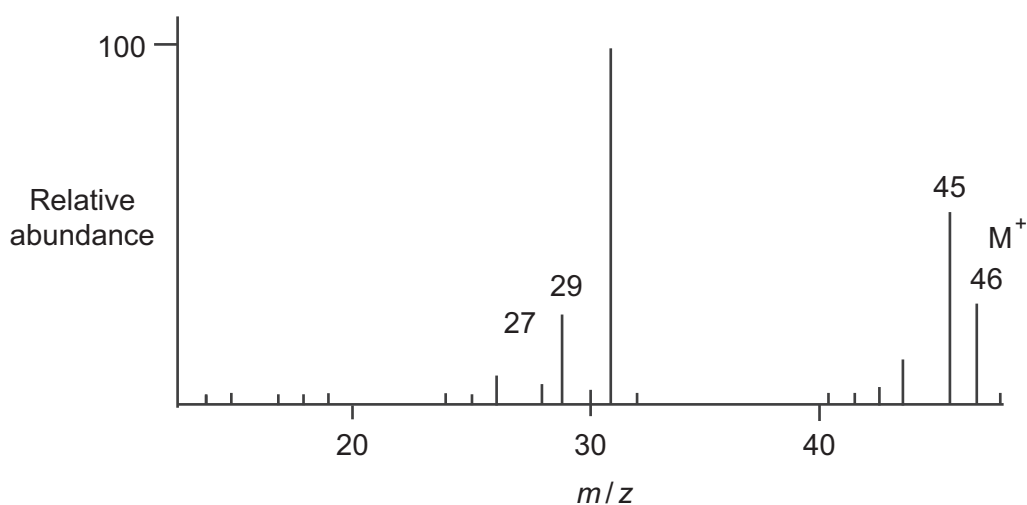


Fig. 7.3.

- (i) Write the formulae of the particles with the m/z value shown.

$m/z = 29$, corresponding particle

$m/z = 45$, corresponding particle [2]

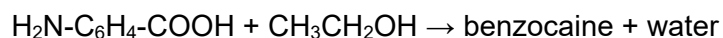
- (ii) Write the meaning of the M^+ peak in mass spectrometry.

..... [1]

[Total 9]

- 8 Benzocaine is a local anesthetic, which means that it makes only small areas of the body insensitive to pain.

The preparation of benzocaine by the esterification of 4-aminobenzoic acid with ethanol is shown.



- (a) State conditions for the reaction above.

..... [2]

- (b) Write the structural formula for benzocaine.

[1]

- (c) (i) Name all the functional groups in benzocaine.

.....
..... [2]

- (ii) Give the systematic name of benzocaine.

..... [1]

- (d) (i) Write down the structural formulae for the products of the reaction between benzocaine and hot aqueous sodium hydroxide.

[1]

- (ii) State the advantages of using sodium hydroxide in the hydrolysis of esters compared to using dilute acid.

.....
.....
.....
..... [2]

[Total: 9]

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9 Water dissociates to form hydroxonium ions and hydroxide ions.

- (a) (i) Write down the equation that shows an acid-base reaction between two water molecules.

[1]

- (ii) Complete the sentences to define the terms Bronsted-Lowry acid and Bronsted-Lowry base.

A Bronsted-Lowry acid is a

A Bronsted-Lowry base is a [2]

- (iii) A Bronsted-Lowry acid has a conjugate base.

State the conjugate base of water.

..... [1]

- (b) (i) Ethanoic acid, CH_3COOH , is a weak acid.

Write the equation for the dissociation of ethanoic acid.

[1]

- (ii) Define the term a weak acid.

..... [1]

- (c) The acid dissociation constant, K_a , for ethanoic acid is $1.8 \times 10^{-5} \text{ mol dm}^{-3}$.

Write an expression for K_a .

Calculate both the $\text{p}K_a$ and the pH of $0.10 \times 10^{-3} \text{ mol dm}^{-3}$ ethanoic acid.

Give your answers to **two** decimal places.

$\text{p}K_a =$

pH = [4]

(d) Calculate the pH of a solution of 0.10 mol dm^{-3} sulfuric acid.

Give your answers to **two** decimal places.

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[3]

[Total 13]

10 Reaction kinetics is the study of the rates of chemical reactions.

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(a) Explain the effect of the surface area of reactants on reaction rates.

.....

.....

.....

..... [3]

(b) The results of rate measurements of the reaction between **A**, **B**, and **C** compounds are given in the table below.

Experiment	Concentration / mol dm ⁻³			Rate of loss of A / mol dm ⁻³ s ⁻¹
	A	B	C	
1	0.010	0.020	0.005	1.4×10^{-6}
2	0.010	0.010	0.005	7.0×10^{-7}
3	0.020	0.020	0.005	1.4×10^{-6}
4	0.020	0.020	0.015	4.2×10^{-6}

(i) Deduce the order of reaction with respect to **A**, **B** and **C**, and the overall order of the reaction.

order of reaction with respect to A

order of reaction with respect to B

order of reaction with respect to C

overall order of reaction [3]

(ii) Write the rate equation for the reaction.

..... [1]

(iii) Calculate a value for the rate constant including units.

rate constant = unit [2]

[Total 9]

11 Amines are derivatives of ammonia.

A primary amine can be prepared from the reaction of *2-bromo-2-methylpropane* and *ammonia*.

(a) Write a balanced chemical equation for this reaction.

All organic compounds should be shown as their structural formulae.

..... [1]

(b) (i) Explain why ammonia should be in **excess** in the reaction.

.....

 [1]

(ii) State the **other** reaction conditions.

..... [1]

(c) Give the name of this type of reaction.

..... [1]

(d) Draw the mechanism for the reaction.

[3]

[Total 7]

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